PART A: THEORY & DEFINITIONS

1. TYPES OF DATA

Numerical Data

Numerical data represents values that can be measured and expressed in numbers. Arithmetic operations such as addition, subtraction, mean, and variance can be performed on numerical data.

Examples from Dataset:

- **study_hours**
- **attendance**
- **previous_test_score**

These columns contain measurable numeric values.

Categorical Data

Categorical data represents labels, categories, or groups rather than numerical values.

Examples from Dataset:

- **group_discussion (Yes/No)**
- **final_exam_pass (Pass/Fail)**

These values represent categories instead of measurable quantities.

TYPES OF STATISTICS

Descriptive Statistics

Descriptive statistics are used to summarize, organize, and describe data.

It helps in understanding:

- **average values,**
- **spread of data,**
- **distribution patterns.**

Examples:

- **Mean**
- **Median**
- **Mode**

- **Standard Deviation**
- **Charts and Graphs**

In this project, descriptive statistics help analyze attendance, study hours, and test scores.

Inferential Statistics

Inferential statistics are used to make predictions or conclusions about a population based on sample data.

It helps:

- **predict outcomes,**
- **test hypotheses,**
- **estimate probabilities.**

Examples:

- **Probability analysis**
- **Bayes theorem**
- **Predictions based on attendance**

WHAT IS DESCRIPTIVE STATISTICS?

Descriptive statistics is the branch of statistics used to summarize and present data in a meaningful way.

It simplifies large datasets using:

- **tables,**
- **graphs,**
- **averages,**
- **measures of dispersion.**

In this project, descriptive statistics helps:

- **understand student performance,**
- **analyze attendance patterns,**
- **compare pass and fail data.**

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2. DIFFERENCE BETWEEN MEAN, MEDIAN, MODE

Mean	Median	Mode
Average of all values	Middle value of ordered data	Most frequently occurring value
Affected by extreme values	Not highly affected by outliers	Represents most common observation
Calculated by sum ÷ total values	Found by arranging data	Found by frequency

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Mean

Mean is the arithmetic average of data.

Formula:

$$\text{Mean} = \frac{\sum x}{n}$$

Where:

- **Σx = sum of all values**
- **n = total number of observations**

Purpose:
Used to find the average study hours, attendance, or scores.

Median

Median is the middle value when data is arranged in ascending or descending order.

If the number of observations is even:

- **median is average of two middle values.**

Purpose:
Used when data contains outliers.

Mode

Mode is the value occurring most frequently in the dataset.

Purpose:
Used to identify most common observation.

DIFFERENCE BETWEEN RANGE, VARIANCE, STANDARD DEVIATION

Range	Variance	Standard Deviation
Difference between maximum and minimum values	Average squared deviation from mean	Square root of variance
Measures spread	Measures variability	Measures dispersion

Range

Range measures the spread between highest and lowest values.

Formula:

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Variance

Variance measures how far data values are spread from the mean.

Formula:

$$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$$

Where:

- **x = individual values**
- **μ = mean**
- **N = total observations**

Standard Deviation

Standard deviation measures average dispersion of data from the mean.

Formula:

$$\sigma = \sqrt{\sigma^2}$$

Purpose:
Used to understand consistency of student performance.

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3. GAUSSIAN DISTRIBUTION

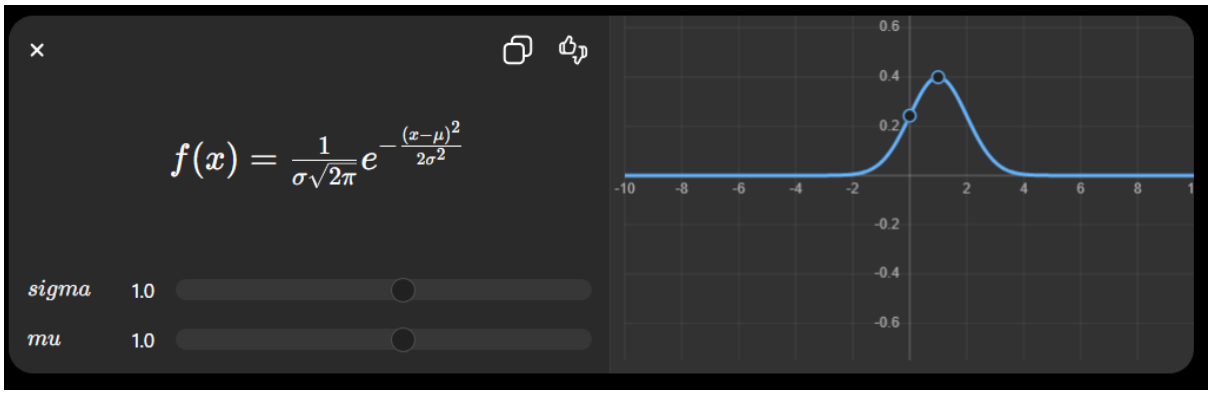
Definition

Gaussian distribution, also called Normal Distribution, is a symmetric bell-shaped distribution where most values lie near the mean.

Characteristics:

- **Mean = Median = Mode**
- **Symmetrical shape**
- **Bell curve**

Formula:



Purpose:
Used to analyze natural distributions such as marks and attendance.

LOG NORMAL DISTRIBUTION

Definition

A log normal distribution occurs when logarithm of variable values follows normal distribution.

Characteristics:

- **Right-skewed distribution**
- **Values cannot be negative**

Used in:

- **financial analysis,**
- **growth data,**
- **income distributions.**

3-SIGMA RULE / EMPIRICAL RULE

Definition

The empirical rule explains data spread in a normal distribution.

Rules:

- **68% data lies within 1 standard deviation**
- **95% data lies within 2 standard deviations**
- **99.7% data lies within 3 standard deviations**

Formula:

$$\mathbf{68\% - 95\% - 99.7\%}\ \boldsymbol{Rule}$$

Purpose:
Used to identify outliers and data spread.

PERCENTILES

Definition

Percentiles divide data into 100 equal parts.

Example:

- **90th percentile means value greater than 90% of observations.**

Purpose:
Used to compare student performance rankings.

QUARTILES

Definition

Quartiles divide data into four equal parts.

Types:

- **Q1 = First Quartile**
- **Q2 = Median**
- **Q3 = Third Quartile**

Purpose:
Used to analyze data distribution.

FIVE NUMBER SUMMARY

Definition

Five number summary describes dataset distribution using five values:

1. **Minimum**
2. **First Quartile (Q1)**
3. **Median (Q2)**
4. **Third Quartile (Q3)**
5. **Maximum**

Purpose:
Used in box plot analysis.

SKEWNESS

Definition

Skewness measures asymmetry of data distribution.

Types:

- **Positive Skewness → tail on right side**
- **Negative Skewness → tail on left side**
- **Zero Skewness → symmetric distribution**

Purpose:
Used to understand distribution shape.

KURTOSIS

Definition

Kurtosis measures peakedness or flatness of distribution.

Types:

- **High kurtosis → sharp peak**
- **Low kurtosis → flat distribution**

Purpose:
Used to identify distribution characteristics and outliers.